Project Title

Phishing-Website-URL-Detection-Analysis Using Python

1. Overview

Phishing is a common cybersecurity threat in which malicious websites attempt to deceive users and collect sensitive information. This project focuses on analyzing phishing and legitimate URLs to identify patterns and characteristics associated with suspicious websites.

The project uses the PhiUSIIL Phishing URL Dataset obtained from the UCI Machine Learning Repository. The original dataset contains 235,795 records and 55 columns, including URL structure, domain information, HTTPS details, and website content features.

Python is used to perform data analysis, explore URL characteristics, identify patterns, and visualize meaningful insights from the dataset.

2. Objectives of the Study

- To understand the distribution of phishing and legitimate URLs.
- To identify missing values and duplicate records in the dataset.
- To clean and prepare the dataset for analysis.
- To analyze URL length, domain length, subdomains, digits, and special characters.
- To examine HTTPS usage and domain-related characteristics.
- To study website features such as titles, favicons, descriptions, and password fields.
- To perform statistical analysis and explore relationships between dataset features.
- To create visualizations to identify patterns associated with phishing and legitimate URLs.
- To derive meaningful insights and recommendations from the analysis.

3. Tools and Technologies Used

Python: Used as the primary programming language for data analysis.

Pandas: Used for loading, cleaning, organizing, and manipulating the dataset.

NumPy: Used for numerical operations and calculations.

Matplotlib: Used to create charts and visualizations.

Seaborn: Used for statistical visualizations and feature relationship analysis.

Google Colab: Used as the cloud-based environment to write and execute Python code.

4. Conclusion

This project aims to provide a better understanding of the characteristics associated with phishing and legitimate URLs through data analysis. By examining URL structures, domain information, HTTPS usage, and website content features, the project seeks to identify meaningful patterns in the dataset.

Using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, the analysis will include data preparation, statistical summaries, exploratory data analysis, and visualizations.

The final findings will help explain differences between phishing and legitimate URLs and provide useful insights into website security characteristics. The analysis is intended for educational and analytical purposes and does not independently guarantee whether a website is safe or malicious.

Dataset Source: UCI Machine Learning Repository — PhiUSIIL Phishing URL Dataset

https://archive.ics.uci.edu/dataset/967/phiusiil+phishing+url+dataset